



B.M.S. COLLEGE OF ENGINEERING **Bengaluru-560019.**

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Visvesvaraya Technological University, Belgaum



Project Report on
**“LoRa-Based Distributed Sensor Network for Real-Time Crowd
Instability and Stampede Risk Detection”**

Submitted in partial fulfilment of the requirement for completion of
PROJECT WORK 1 [23EC6PWPJ1]

Submitted by

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CERTIFICATE

This is to certify that the Project Work 1 entitled “**LoRa-Based Distributed Sensor Network for Real-Time Crowd Instability and Stampede Risk Detection**” is a bonafide work carried out by **Adithya C(1BM23EC013)**, **Arnav Sharma(1BM23EC038)**, **Ayush Deep(1BM23EC045)** and **Baibhav Giri (1BM23EC049)** submitted in partial fulfilment of the requirement for completion of PROJECT WORK 1 [23EC6PWPJ1] of Bachelor of Engineering in Electronics and Communication during the academic year 2025-26. The Project Work 1 report has been approved as it satisfies the academic requirements.

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DECLARATION

We, **Adithya C(1BM23EC013)**, **Arnav Sharma(1BM23EC038)**, **Ayush Deep(1BM23EC045)** and **Baibhav Giri (1BM23EC049)** hereby declare that the Project Work 1 entitled '**LoRa-Based Distributed Sensor Network for Real-Time Crowd Instability and Stampede Risk Detection**' is a bonafide work and has been carried out by us under the guidance of **Dr. Hemavathi D**, Assistant Professor, Department of Electronics and Communication Engineering, BMS College of Engineering, Bengaluru submitted in partial fulfilment of the requirement for completion of PROJECT WORK 1 [23EC6PWPJ1] of Bachelor of Engineering in Electronics and Communication during the academic year 2025 - 26. The Project Work 1 report has been approved as it satisfies the academic requirements in Electronics and Communication engineering, Visvesvaraya Technological University, Belagavi, during the academic year 2025 - 26. We further declare that, to the best of our knowledge and belief, this project work 1 has not been submitted either in part or in full to any other university.

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ABSTRACT

Crowd stampedes are among the most dangerous public safety disasters occurring in high-density environments such as religious gatherings, railway stations, stadiums, and public events. Traditional monitoring systems based on CCTV surveillance suffer from limitations including high latency, dependence on continuous human monitoring, privacy concerns, and a critical dependency on internet infrastructure. This work presents the design and implementation of a distributed, internet-independent LoRa-based sensor network for real-time crowd instability and stampede risk detection using multi-sensor fusion and adaptive threshold analysis.

The system architecture consists of multiple edge-computing sensor nodes equipped with ESP32 microcontrollers, load cells (HX711), accelerometers (ADXL345), and microphone modules (MAX9814). These nodes continuously monitor physical crowd parameters including localized pressure buildup, ground vibration energy, and ambient sound intensity. To ensure data integrity, each node performs localized signal processing and feature extraction, utilizing moving-average filters to suppress transient environmental noise.

A central contribution of this research is the development of a unified Crowd Instability Index (CII). This index fuses normalized sensor data and their respective rates of change using weighted coefficients to provide a comprehensive risk score. To minimize false alarms caused by isolated transient disturbances, the system employs adaptive statistical thresholding ($\mu + k$) and sliding-window temporal analysis to identify sustained abnormal crowd behaviour. Data is transmitted via SX1278 LoRa modules, providing long-range communication (up to 5km) with low power consumption, ensuring the system remains operational even during network grid failures.

The implemented prototype successfully demonstrated real-time sensor-data acquisition, distributed anomaly detection, and adaptive alert generation under controlled crowd-instability simulation scenarios. Experimental validation showed reliable differentiation between four distinct states: normal crowd conditions ($CII = 0.15$), high-density states, panic movement ($CII = 0.54$), and critical crowd-surge conditions ($CII > 0.60$). The proposed architecture operates entirely without camera surveillance or cloud infrastructure, offering a scalable, low-power, and privacy-preserving solution for public safety monitoring in diverse infrastructure-poor environments.

Through elimination of the need for video surveillance or cloud-based processing, the proposed architecture provides a scalable, cost-effective, and privacy-preserving solution for public safety monitoring. This research establishes a foundation for "Infrastructure-as-a-Sensor" (IaaS), offering a modular blueprint for future smart-city safety frameworks that prioritizes rapid, autonomous response.

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List of abbreviations

Abbreviation	Meaning
LoRa	Long Range
CII	Crowd Instability Index
RMS	Root Mean Square
MCU	Microcontroller Unit
FFT	Fast Fourier Transform
IoT	Internet of Things
ADC	Analog to Digital Converter

Chapter 1:

Introduction

In the modern era of rapid urbanization and global connectivity, mass gatherings at religious festivals, sporting arenas, transit hubs, and political rallies have become increasingly frequent. While these events signify social cohesion, they also present a catastrophic risk: the crowd stampede. Historical data suggests that crowd-related disasters are rarely the result of a single "panic" event, but rather a cumulative failure of situational awareness and physical space management. When crowd density exceeds a critical threshold—typically four to six persons per square meter—individual movement becomes impossible, and the crowd begins to behave like a fluid. In this state, "shockwaves" of pressure can travel through the mass, leading to structural failures of barricades or collective falls that trigger a deadly chain reaction. Currently, the most prevalent method for crowd management is Closed-Circuit Television (CCTV) paired with manual human oversight. However, this approach faces several systemic vulnerabilities. First, human operators are prone to fatigue and "vigilance decrement," often failing to spot the subtle precursors of a surge in a sea of visual data. Second, computer vision (CV) algorithms, while advancing, require significant computational power (GPU-intensive) and high-bandwidth internet connectivity, which are often the first to fail during a large-scale emergency. Furthermore, the invasive nature of video surveillance raises significant ethical and privacy concerns. There is, therefore, a critical need for a non-intrusive, decentralized, and internet-independent system that can monitor the "physicality" of a crowd rather than its visual representation. The Physics of Crowd Instability Academic research into crowd dynamics (specifically the "Social Force Model") indicates that abnormal physical indicators emerge well before a critical collapse occurs.

These precursors include:

- **Localized Pressure Accumulation:** Physical force exerted by individuals against fixed structures or each other.
- **Kinetic Vibration:** High-frequency vibrations in the ground or flooring caused by erratic, non-laminar movement or rhythmic surging.
- **Acoustic Anomalies:** Sudden spikes in sound intensity and frequency shifts that characterize panic or distress.
- **Temporal Surges:** Rapid changes in density over a short time window. By shifting the focus from *visual monitoring* to *physical sensing*, it is possible to detect the "stress" of a crowd in real-time, even in low-light conditions or obstructed environments where cameras would fail.

This project proposes a distributed, LoRa-based sensor network designed to provide an early-warning system for crowd instability. The system leverages "Edge Intelligence," where the heavy lifting of data analysis is performed at the local node level. Each node integrates a specialized sensor suite comprising load cells, accelerometers, and acoustic sensors to capture the multi-modal physical state of the environment.

The core objectives of this work are:

1. To develop a decentralized sensing architecture that operates independently of the power grid and the internet.
2. To implement a multi-sensor fusion algorithm, known as the Crowd Instability Index (CII), to quantify risk levels accurately.
3. To utilize Long Range (LoRa) communication protocols to ensure data transmission over vast, congested areas with minimal power consumption.
4. To create an adaptive thresholding mechanism that distinguishes between normal high-density activity and genuine life-threatening surges.

1.1 Area of Application

The proposed system is designed for deployment in high-density public environments where sudden crowd instability and stampede situations may occur.

Applications include:

- Religious gatherings and festivals
- Railway stations and metro platforms
- Stadiums and sports arenas
- Political rallies and public events
- Shopping malls and exhibition centres
- Emergency evacuation monitoring systems

1.2 Existing System Limitations

Existing crowd-monitoring systems are predominantly based on centralized CCTV surveillance and internet-connected analytics platforms.

These systems suffer from several limitations:

- Dependence on continuous human supervision
- Internet and cloud dependency
- High computational requirements
- Privacy concerns due to camera surveillance
- Delayed identification of physical crowd instability

1.3 Motivation & Reasons for the project

The increasing occurrence of crowd-related disasters highlights the need for decentralized, privacy-preserving, and real-time monitoring systems capable of identifying physical indicators of crowd instability before critical collapse occurs.

The proposed project aims to develop a lightweight distributed sensing architecture capable of detecting abnormal crowd conditions using locally processed sensor information and long-range wireless communication.

1.4 System Philosophy

The proposed system follows a distributed edge-sensing architecture in which multiple sensor nodes continuously monitor physical crowd parameters such as pressure buildup, vibration energy, and sound intensity.

Instead of continuously transmitting raw sensor data, each node performs localized signal processing and anomaly estimation before transmitting summarized risk information using LoRa communication. This architecture improves scalability, reduces communication overhead, and enables rapid deployment in high-density public environments.

Chapter 2

Literature survey

2.1 Historical Context and the Evolution of Crowd Dynamics

The scientific study of crowd disasters has shifted from qualitative observation to quantitative physical modelling. Initial studies by Helbing et al. [1] and Still [2] established that crowd "panic" is often a misnomer; the true cause of fatalities is typically "crowd turbulence." This is a state where physical density becomes so high that forces are transmitted through human bodies as longitudinal waves. Recent work in *Accident Analysis & Prevention* (2025) suggests that monitoring these physical force-chains is more critical than monitoring individual behaviour, as the collective "fluidized" state of a crowd is what precedes a stampede [Bib-6].

2.2 Technical Limitations of Computer Vision (CV) in Emergency Contexts

While modern CV techniques—including CNN-LSTMs [Bib-1] and Video Swin Transformers [Bib-3]—provide high-resolution density maps, they face "Real-World Deployment Constraints":

- Occlusion and Lighting: Cameras fail in low-light environments or when smoke/dust is present, which are common in emergency scenarios.
- Computational Latency: High-fidelity video analytics require GPU-heavy centralized servers, leading to significant delays between event detection and alert generation [16].
- Network Vulnerability: Centralized systems depend on high-speed internet. During mass gatherings, cellular towers often reach peak capacity or fail entirely, rendering cloud-based safety systems useless [Bib-2].
- Privacy Ethics: As highlighted in *Discover Computing* (2024), the increasing public resistance to facial recognition and constant visual monitoring necessitates the development of "Zero-Visibility" sensing [Bib-2, Bib-5].

2.3 The Physics of Physical Precursors: Pressure and Vibration

Research into the "Social Force Model" indicates that the transition from a safe crowd to a dangerous one is marked by specific physical anomalies.

- Pressure Accumulation: When individuals are packed at densities exceeding 5 people per M^2 the force exerted between bodies can reach several thousand Newtons. This pressure is most dangerous at "bottlenecks" like exits or narrow corridors [1, 6].
- Kinetic Energy and Vibration: Ground-plane vibrations provide a non-intrusive way to measure crowd movement energy. Kim et al. [17] demonstrated that abnormal movement patterns, such as "rhythmic surging," create distinct spectral signatures in the 1–10 Hz range, which can be captured by MEMS accelerometers like the ADXL345.

2.4 Multi-Modal Sensor Fusion and The Crowd Instability Index (CII)

The limitation of single-sensor systems is the high rate of false positives. For instance, a single loud noise might trigger an acoustic sensor, but not a pressure sensor. Current research emphasizes "Multi-Modal Fusion," where data from different physical domains (pressure, vibration, sound) are combined [15]. By assigning weights to these parameters—localized force (P), vibration energy (V), and sound intensity (S)—researchers can compute a unified Crowd Instability Index (CII). This mathematical approach allows for a "graduated alert" system rather than a binary "on/off" alarm, providing security personnel with early warnings before a critical threshold is crossed [9, 12].

2.5 Edge Intelligence and Decentralized Processing Paradigms

The "Edge-at-the-Source" paradigm involves moving the intelligence from the cloud to the sensor node itself. Utilizing microcontrollers like the ESP32 allows for:

- Localized Signal Processing: Applying Moving-Average Filters and Fast Fourier Transforms (FFT) directly on the node to filter out transient noise [10].
- Bandwidth Efficiency: Transmitting only a 1-byte "Risk Level" instead of raw high-frequency sensor data, which is crucial for low-bandwidth protocols like LoRa [18].
- Fault Tolerance: In a distributed network, the failure of one "Edge Node" does not compromise the entire system, providing a layer of redundancy missing in centralized architectures [8, 13].

2.6 LoRa (Long Range) vs. Traditional Wireless Protocols

A critical area of recent study is the reliability of wireless communication in congested environments. Standard protocols such as Wi-Fi and Bluetooth operate in the 2.4 GHz band, which suffers from extreme absorption by human bodies (which are primarily water).

- Sub-GHz Advantages: LoRa operates at lower frequencies (868/915 MHz), allowing for much greater penetration through human masses and concrete structures [4, 5].
- Low-Power Wide-Area Network (LPWAN): Studies in the *IoT Journal* (2025) confirm that LoRa can maintain a "Link Budget" sufficient for 2–5km range even in dense urban canyons [Bib-8].
- Network Scalability: LoRaWAN's "Star-of-Stars" topology allows a single gateway to support thousands of nodes, making it ideal for vast event spaces like stadiums or pilgrimage sites [14, 20].

2.7 AI-Driven Predictive Analytics and Tiny-ML

The most advanced research for 2026 focuses on "Tiny-ML"—the deployment of machine learning models on microcontrollers with limited RAM. Techniques such as Sparse Feature Tracking [Bib-4] and Tiny-Scale DETR [Bib-7] enable the system to recognize *patterns* of movement rather than just *thresholds*. For example, the system can distinguish between a "rhythmic dance" (low risk) and a "chaotic surge" (high risk) by

analyzing the temporal correlation between distributed nodes [Bib-9]. This "Intelligence-at-the-Edge" approach is the current gold standard for autonomous public safety infrastructure.

2.8 Summary of the Research Gap

While the individual components of crowd sensing (pressure sensors, vibration analysis, and LoRa) are well-documented, the integration of these into a Distributed, Internet-Independent, Multi-Sensor Network remains a significant research gap. Existing solutions are either too computationally expensive (CV-based) or too simplistic (single-threshold sensors). This project bridges the gap by implementing a robust, privacy-preserving framework that utilizes multi-modal sensor fusion and LoRa communication to provide real-time, autonomous safety monitoring in infrastructure-poor environments.

Chapter 3:

Problem Analysis & Solution

3.1 Problem Definition

The management of large-scale public gatherings remains one of the most significant challenges in modern urban safety, particularly in high-density regions where unpredictable human behavior can lead to catastrophic outcomes. Recent incidents underscore the volatility of these environments. For instance, at the M. Chinnaswamy Stadium in Bengaluru, the sheer volume of spectators during high-profile matches creates immense bottlenecks at entry and exit gates. In such scenarios, the transition from a "flowing" crowd to a "compacted" state occurs in seconds, often outstripping the reaction time of manual security personnel. Similarly, the Karur stampede during a political gathering highlighted a critical failure in situational awareness; sudden panic or density surges in confined areas can lead to fatalities long before external emergency services are even notified. The fundamental problem lies in the inadequacy of current monitoring paradigms. Traditional systems depend almost exclusively on CCTV surveillance and manual supervision, which suffer from systemic vulnerabilities. These include high response latency, a total dependence on stable internet connectivity—which often collapses during mass surges due to network congestion—and the significant privacy concerns associated with constant video recording. Furthermore, these systems are centralized, meaning a single point of failure in the server or power grid can blind the entire safety infrastructure. As seen in the Karur incident, visual monitoring is often obstructed by the density of the crowd itself, preventing security from seeing the physical "pressure waves" that precede a crush.

Consequently, there is a critical need for a decentralized, sensor-based monitoring system that focuses on the physical markers of crowd instability rather than visual cues. Such a system must be built on a low-power architecture capable of long-range, internet-independent wireless communication to ensure it remains operational when cellular networks fail. It must provide real-time anomaly detection through adaptive thresholding, distinguishing between the rhythmic movement of a stadium cheer and the chaotic, sustained pressure of a life-threatening surge.

To address these challenges, the system must continuously monitor physical crowd indicators—such as localized pressure, ground vibration, and acoustic spikes—and transmit summarized risk data reliably over vast distances. By performing localized edge-processing, the architecture must detect abnormal surge conditions at the source and trigger graduated alerts with minimal delay. Ultimately, this project seeks to bridge the gap between physical dynamics and digital safety, providing a scalable, privacy-preserving framework for public safety in high-risk, infrastructure-poor environments.

3.2 System Overview

The system consists of:

Sensor Node → ESP32 → LoRa Module → LoRa Gateway → Processing Unit →
Alert System

Sensor Node Design

Each distributed sensor node consists of:

- ESP32 Microcontroller
- SX1278 LoRa Communication Module
- 5kg Load Cell with HX711 Amplifier
- ADXL345 3-Axis Accelerometer
- MAX9814 Microphone Module

The sensor node continuously monitors physical crowd parameters and performs localized edge processing before transmission.

The load cell measures localized force buildup associated with crowd compression. The ADXL345 accelerometer captures abnormal movement patterns and vibration energy generated during crowd surges. The microphone module detects sudden sound-energy spikes associated with panic conditions.

Each node performs:

Sensor acquisition

- Moving-average filtering
- RMS vibration computation
- Signal normalization
- Adaptive threshold analysis

- Crowd Instability Index computation
- LoRa packet transmission

3.2 Advantages of Proposed System

The proposed crowd instability and stampede risk detection system offers several advantages over conventional crowd-monitoring approaches that mainly rely on CCTV surveillance and centralized monitoring infrastructure.

1. Real-Time Physical Crowd Monitoring:The system continuously monitors physical crowd-behavior indicators such as pressure buildup, vibration intensity, and abnormal sound levels using distributed sensor nodes. This enables early identification of unstable crowd conditions before a critical crowd surge or stampede occurs.

2. Distributed Edge Processing:Each sensor node performs localized signal acquisition, filtering, feature extraction, and Crowd Instability Index (CII) computation. Since preliminary processing is performed at the sensor level itself, the system reduces centralized computational load and improves response speed.

3. Low-Latency Alert Generation:The proposed architecture minimizes communication overhead by transmitting only summarized risk information instead of continuous raw sensor streams. This allows rapid detection and faster emergency alert generation during abnormal crowd conditions.

4. Long-Range Wireless Communication:LoRa communication technology enables reliable long-range wireless transmission with very low power consumption. The system can therefore operate efficiently in large public areas such as railway stations, stadiums, temples, and political gatherings without requiring extensive network infrastructure.

5. Low Power Consumption:The ESP32 microcontroller and LoRa communication module are optimized for low-power embedded operation. The event-driven communication mechanism further reduces unnecessary transmissions, making the system suitable for battery-powered and portable deployments.

6. Privacy-Preserving Architecture:Unlike traditional CCTV-based systems, the proposed system does not rely on cameras or video analytics. Only physical crowd parameters are monitored, thereby avoiding privacy concerns associated with image acquisition and facial surveillance.

7. Adaptive Threshold-Based Detection:The system uses adaptive statistical thresholding instead of fixed thresholds. Dynamic thresholds are continuously updated based on moving averages and standard deviation values of sensor readings. This improves robustness under varying environmental conditions and reduces false alarms.

8. Multi-Parameter Sensor Fusion:The proposed approach combines pressure sensing, vibration sensing, and sound-intensity monitoring to improve anomaly detection accuracy. Sensor fusion provides better reliability compared to systems depending on a single sensing parameter.

9. Scalability and Modular Expansion:Additional distributed sensor nodes can easily be integrated into the network without major architectural changes. This allows scalable deployment for large public events and smart-city crowd-monitoring applications.

10. Internet-Independent Operation:The system does not require continuous internet connectivity or cloud processing for operation. Localized processing and LoRa-based communication allow reliable operation even in environments with poor network infrastructure.

11. Early Warning Capability:The system detects gradual increases in crowd instability indicators such as pressure growth rate and vibration energy before catastrophic conditions occur. This enables early warning generation and supports preventive crowd-management strategies.

12. Cost-Effective Implementation:The proposed prototype uses commercially available embedded hardware components such as ESP32, SX1278 LoRa modules, HX711 amplifiers, and MEMS sensors. This reduces implementation cost while maintaining reliable system performance.

13. Suitability for Smart Public Safety Systems:The proposed architecture can be integrated into future smart-city infrastructure for automated crowd-risk monitoring, emergency management, and disaster-prevention applications.

Chapter 4:

Methodology & Implementation

4.1 System Architecture

The proposed system follows a distributed sensing architecture in which multiple sensor nodes continuously monitor physical crowd parameters and transmit summarized risk information through LoRa communication.

Each sensor node performs localized signal acquisition, signal processing, feature extraction, and anomaly estimation before transmitting risk information to the monitoring unit.

The overall system consists of:

- Distributed sensor nodes
- LoRa communication network
- Monitoring and alert system
- Local signal-processing unit

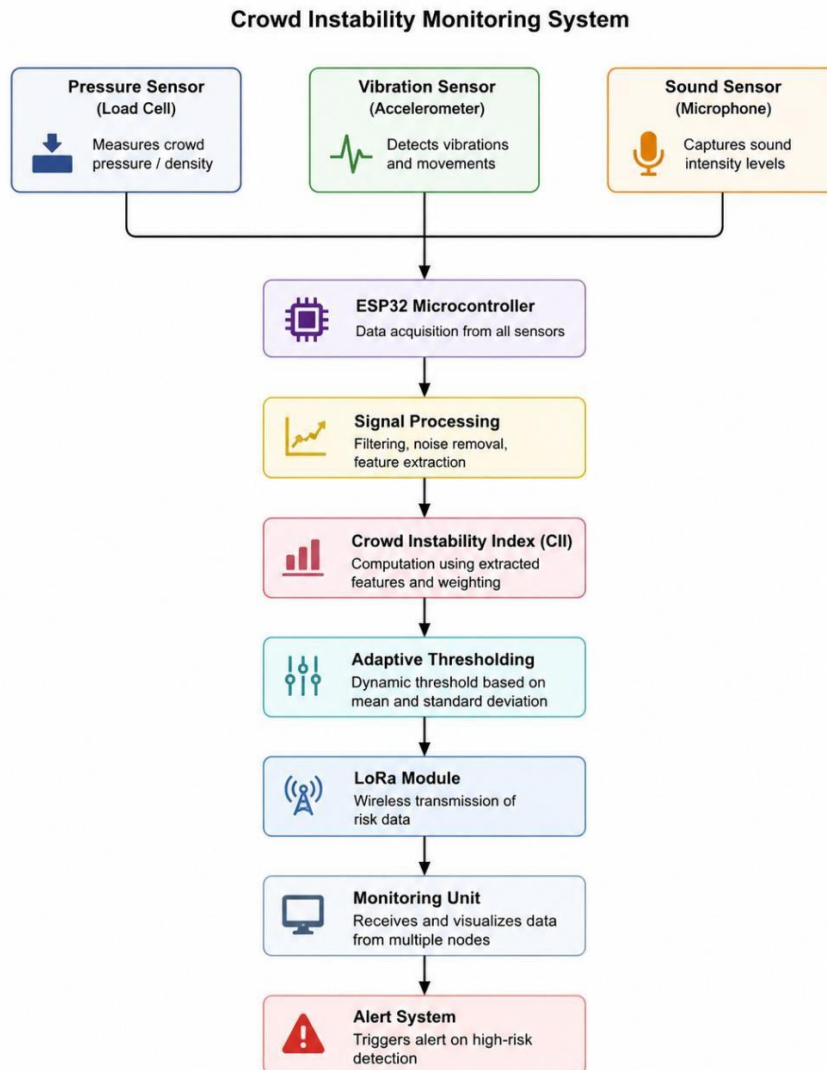


Figure 4.1:Overall System Architecture

4.2 Sensor Node Structure

Each sensor node consists of sensing, processing, and communication units integrated into a localized edge-processing architecture.

The sensor node includes:

- ESP32 microcontroller
- SX1278 LoRa communication module
- 5kg load cell with HX711 amplifier
- ADXL345 accelerometer
- MAX9814 microphone module

The ESP32 performs sensor acquisition, filtering, feature extraction, adaptive threshold analysis, and local anomaly estimation before wireless transmission.

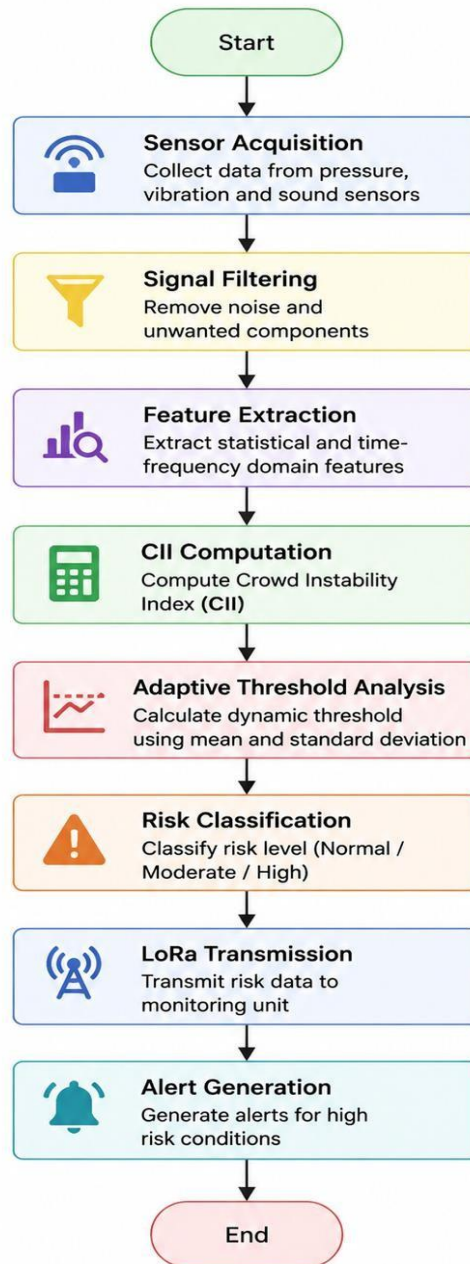


Figure 4.2: Sensor Node Structure

4.3 Sensor Node Design

The sensor node continuously monitors pressure buildup, vibration energy, and sound intensity within the crowd environment.

The load cell measures localized force buildup associated with excessive crowd compression. The ADXL345 accelerometer captures abnormal movement patterns and vibration energy generated during crowd surges. The microphone module detects sudden sound-energy spikes associated with panic conditions.

Each node performs:

- Sensor acquisition
- Signal normalization
- Moving-average filtering
- RMS vibration computation
- Adaptive threshold analysis
- Crowd Instability Index computation
- LoRa packet transmission

4.3.1 Hardware Components

1. ESP32 DEVELOPMENT BOARD

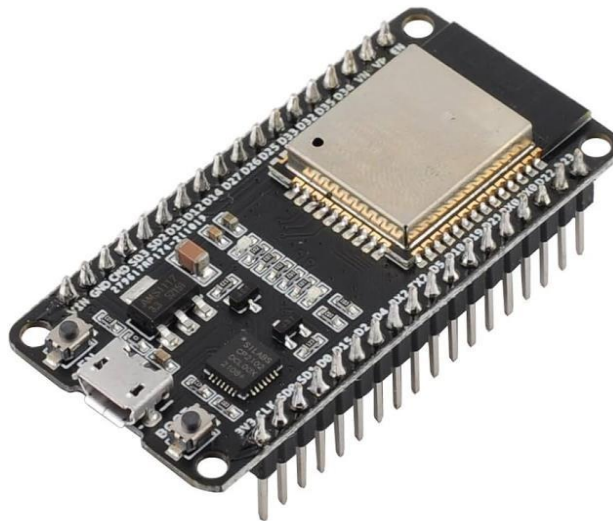


Fig 4.4: ESP32 Development Board

Specifications:

- Microcontroller: ESP32 Dual-Core Xtensa LX6
 - Clock Frequency: Up to 240 MHz
 - Operating Voltage: 3.3V
 - SRAM: 520 KB
 - Flash Memory: 4 MB
 - Communication Interfaces: UART, SPI, I2C, PWM
 - Wireless Features: WiFi and Bluetooth

Role in Overall System:

The ESP32 acts as the primary processing unit of each distributed sensor node. It acquires sensor data from the pressure, vibration, and sound sensors, performs localized signal processing, computes the Crowd Instability Index (CII), and manages LoRa communication with the monitoring unit.

Advantages:

- High processing capability
 - Low power consumption
 - Multiple communication interfaces
 - Compact embedded design

2. SX1278 LORA MODULE

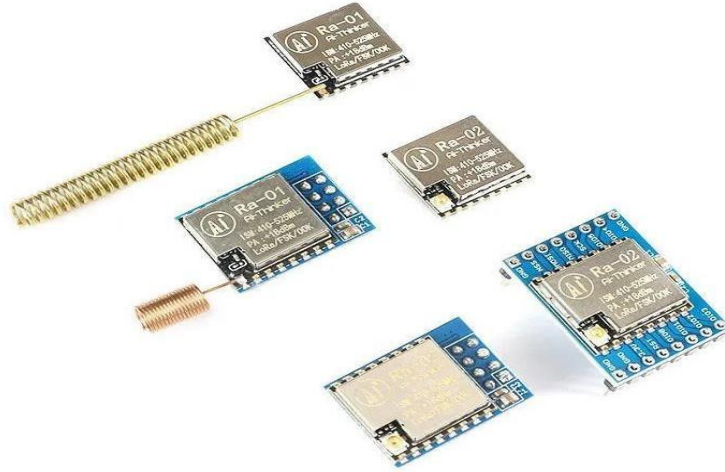


Fig 4.5: SX1278 LoRa Module

Specifications:

- Operating Frequency: 433 MHz
- Communication Range: Up to 5 km (open environment)
- Modulation Technique: LoRa Spread Spectrum
- Operating Voltage: 3.3V
- Interface: SPI
- Low-Power Sleep Current: $< 1 \mu\text{A}$

Role in Proposed System:

The SX1278 LoRa module provides long-range wireless communication between distributed sensor nodes and the monitoring unit. It transmits summarized crowd-risk information with low communication overhead and reduced power consumption.

Advantages:

- Long communication range
- Low power operation
- Reliable wireless transmission
- Internet-independent communication

3. LOAD CELL WITH HX711



Fig 4.6: Load Cell with HX711 Amplifier

Specifications:

Load Cell:

- Maximum Load Capacity: 5 kg
- Output Sensitivity: 1 mV/V
- Material: Aluminum Alloy

HX711 Amplifier:

- ADC Resolution: 24-bit
- Operating Voltage: 2.6V – 5.5V
- Interface: Digital Serial Interface

Role in Proposed System:

The load cell measures localized pressure buildup caused by crowd density and compressive force accumulation. The HX711 amplifier converts low-level analog signals from the load cell into digital values suitable for ESP32 processing.

Advantages:

- High measurement sensitivity
- Accurate pressure monitoring
- Stable digital conversion
- Low-noise amplification

4. ADXL345 ACCELEROMETER

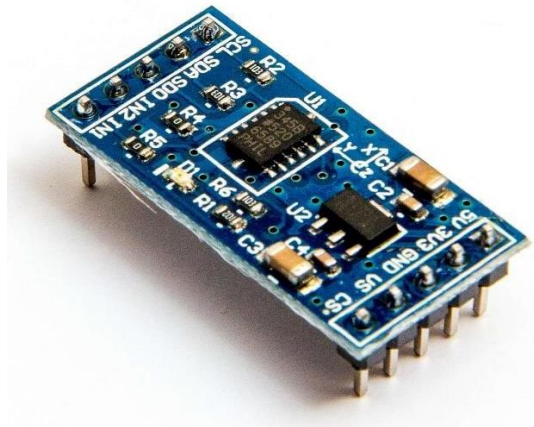


Fig 4.7: ADXL345 Accelerometer

Specifications:

- Operating Voltage: 2.7V – 5.5V
 - Automatic Gain Control (AGC): Yes
 - Amplifier Gain: 40 dB / 50 dB / 60 dB
 - Output Type: Analog Output
 - Low-Noise Microphone Amplifier

Role in Proposed System:

The MAX9814 microphone module detects abnormal sound-intensity spikes generated during panic situations, shouting, or sudden crowd disturbances. The sensor assists in identifying panic-induced crowd instability conditions.

Advantages:

- Automatic gain adjustment
 - High audio sensitivity
 - Low-noise operation
 - Compact analog interface

5. MAX9814 MICROPHONE MODULE



Fig 4.8: MAX9814 Microphone Module

Specifications:

- Operating Voltage: 2.0V – 3.6V
- Interface: I²C / SPI Digital Communication
- Measurement Axes: 3-axis (X, Y, Z)
- Measurement Range: $\pm 2g$, $\pm 4g$, $\pm 8g$, $\pm 16g$
- Resolution: 13-bit
- Low Power Consumption: 23 μA (typical)
- Output Type: Digital Output
- High Sensitivity Motion Detection
- Built-in Free-fall and Activity Detection

Role in Proposed System:

The ADXL345 accelerometer is used to detect sudden vibrations, abnormal movements, and crowd instability conditions. During panic situations or stampede-like events, the sensor measures rapid changes in acceleration and body movement patterns. These motion variations help the system identify unusual crowd behavior and provide early warning signals for potential crowd disasters.

Advantages:

- High sensitivity for motion and vibration detection
- Compact and lightweight module
- Low power consumption
- Supports both I²C and SPI communication
- Accurate 3-axis acceleration measurement
- Suitable for real-time monitoring applications

4.2.2 Transmitter Node Implementation

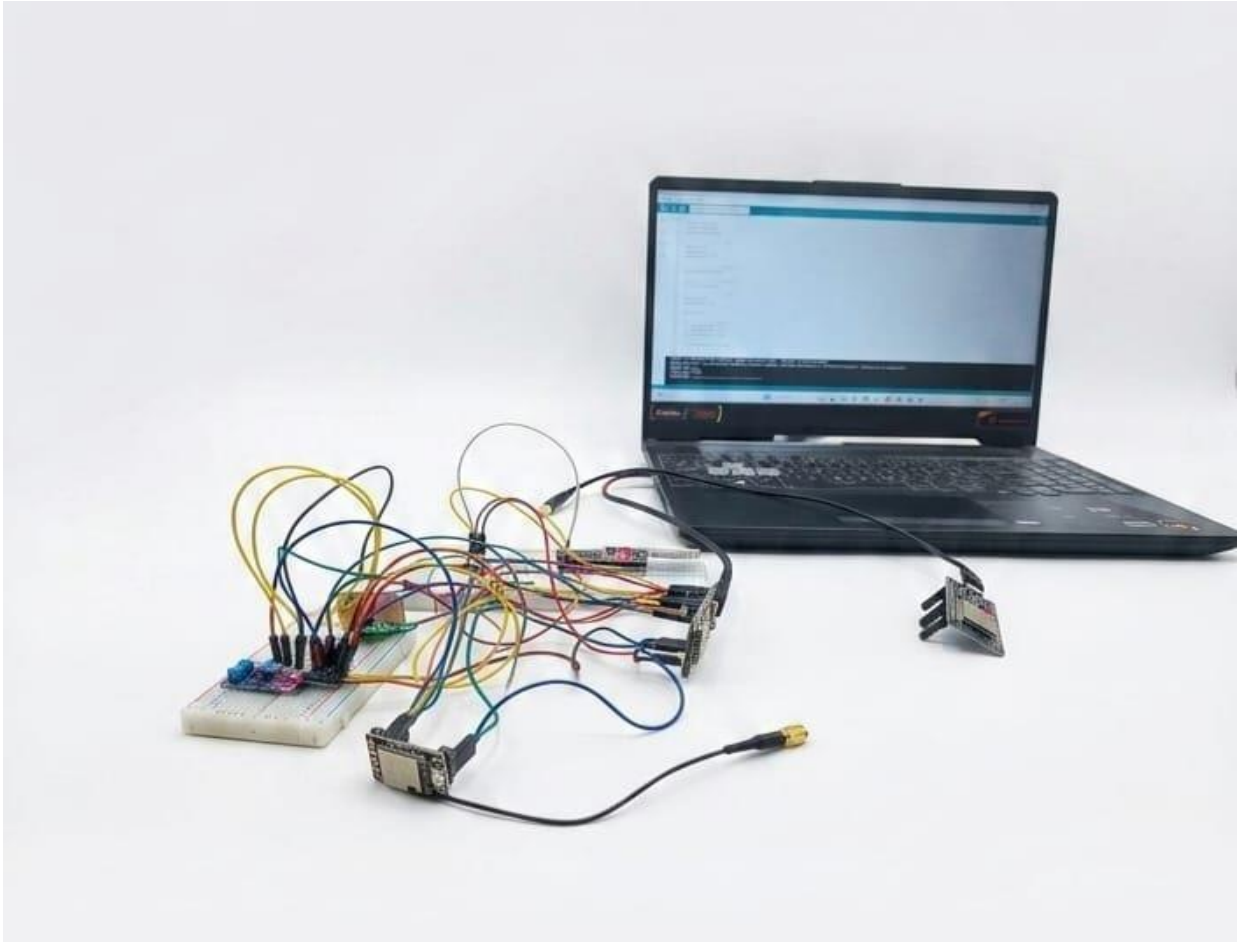


Fig 4.9: Transmitter Sensor Node Prototype

The transmitter node consists of an ESP32 microcontroller integrated with pressure, vibration, and sound sensors along with the SX1278 LoRa communication module.

The node continuously acquires physical crowd-behavior parameters, performs localized signal processing, computes the Crowd Instability Index (CII), and transmits summarized risk information to the monitoring unit through LoRa communication.

The implemented prototype validates distributed sensing and localized edge-processing functionality for real-time crowd instability monitoring.

4.2.3 Receiver Node Implementation

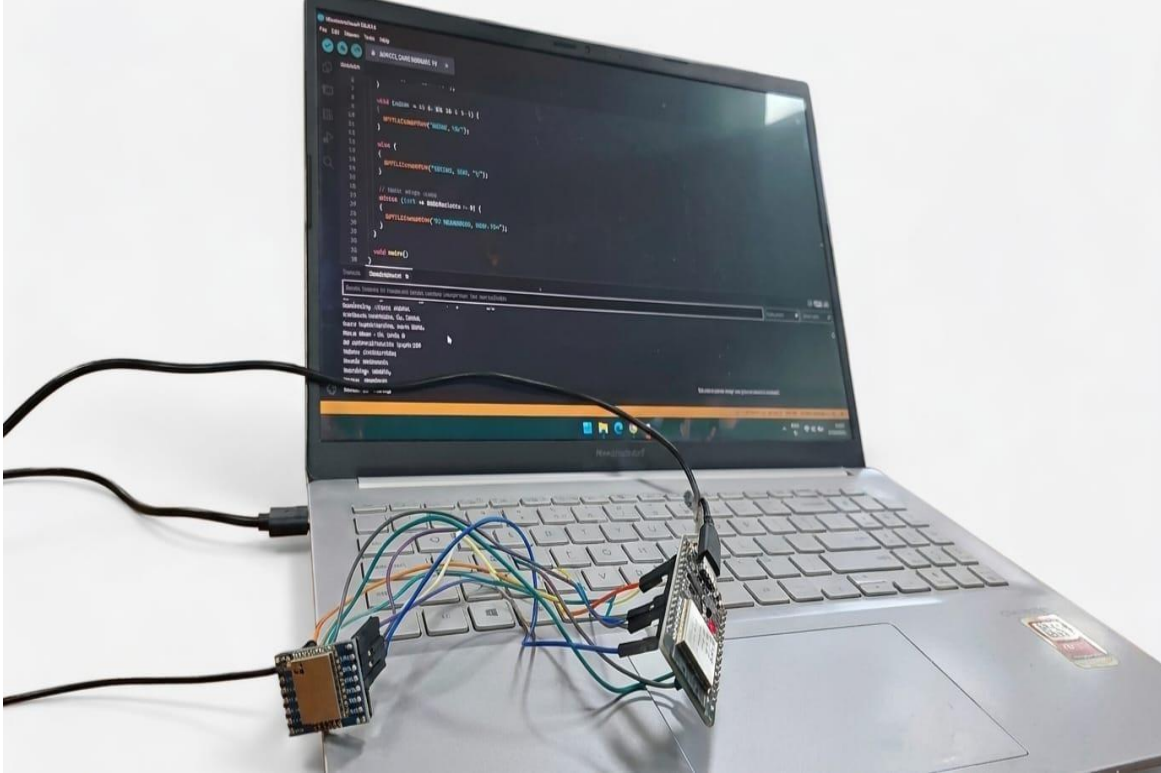


Fig 4.10: Receiver Node Implementation

The receiver unit consists of an ESP32-based LoRa receiver integrated with alert indicators and monitoring interfaces. The receiver node collects transmitted risk information from distributed sensor nodes and generates warning alerts during abnormal crowd conditions. The implemented setup validates long-range wireless communication, real-time risk monitoring, and alert-generation functionality of the proposed system.

4.3 Signal Processing and Feature Extraction

Raw sensor signals are processed locally at each node to extract meaningful crowd-behavior indicators.

The extracted parameters include:

- Pressure value (P)
- Vibration energy (V)
- Sound intensity (S)
- Pressure growth rate (ΔP)
- Vibration growth rate (ΔV)

Moving-average filtering and sliding-window analysis are used to suppress transient noise and detect sustained abnormal crowd behavior.

Table 4.1: Sensor Readings and System Output

Test Case	Pressure (P)	Vibration (V)	Sound (S)	Derivative (D)	Output
Normal Crowd	0.20	0.15	0.18	0.05	LESS CROWDED
Moderate Crowd	0.40	0.35	0.30	0.10	CROWDED
Panic Movement	0.62	0.70	0.58	0.18	CROWD SURGE
Critical Surge	0.88	0.92	0.85	0.32	STAMPEDE RISK

4.4 Crowd Instability Index (CII)

A unified Crowd Instability Index (CII) is computed using fused sensor parameters.

$$CII = w_1P + w_2V + w_3S + w_4\Delta P + w_5\Delta V$$

Where:

- P = normalized pressure value
- V = vibration energy
- S = sound intensity
- ΔP = pressure growth rate
- ΔV = vibration growth rate
- w_1, w_2, w_3, w_4, w_5 = weighting coefficients

The CII provides a quantitative estimate of overall crowd instability.

Table4.2: Crowd Instability Index (CII) Computation

Test Case	Equation Calculation	CII Value	Classification
Normal Crowd	$(0.35 \times 0.20) + (0.25 \times 0.15) + (0.20 \times 0.18) + (0.20 \times 0.05)$	0.1535	LESS CROWDED
Moderate Crowd	$(0.35 \times 0.40) + (0.25 \times 0.35) + (0.20 \times 0.30) + (0.20 \times 0.10)$	0.3075	CROWDED
Panic Movement	$(0.35 \times 0.62) + (0.25 \times 0.70) + (0.20 \times 0.58) + (0.20 \times 0.18)$	0.5440	CROWD SURGE
Critical Surge	$(0.35 \times 0.88) + (0.25 \times 0.92) + (0.20 \times 0.85) + (0.20 \times 0.32)$	0.7720	STAMPEDE RISK

4.5 Adaptive Threshold Detection

The proposed system utilizes adaptive statistical thresholding to identify sustained abnormal crowd conditions.

The adaptive threshold is defined as:

$$Threshold = \mu + k\sigma$$

Where:

- μ = moving average
- σ = moving standard deviation
- k = scaling factor

Risk levels are classified as:

- Normal
- High Risk
- Critical

Adaptive thresholding improves robustness and reduces false alarms caused by environmental noise.

Table 4.3: Adaptive Threshold Analysis

Test Case	CII Value	Threshold Value	Result
Normal Crowd	0.1535	0.60	Safe Condition
Moderate Crowd	0.3075	0.60	Moderate Crowd
Panic Movement	0.5440	0.60	High Crowd Activity
Critical Surge	0.7720	0.60	Emergency Alert Triggered

4.6 Temporal Analysis

Crowd instability develops progressively over time rather than instantaneously. Temporal analysis is performed using sliding-window processing to capture sustained abnormal activity.

The system:

- Monitors sensor behavior over fixed intervals
- Filters transient disturbances
- Detects sustained anomalies
- Tracks rapid parameter growth trends

This improves system reliability and prevents false triggering from short-

duration disturbances.

4.7 Local Decision Making

Each sensor node independently evaluates crowd conditions using locally processed sensor data and the computed Crowd Instability Index.

Instead of continuously transmitting raw sensor streams, only summarized risk information is transmitted through LoRa communication.

This approach provides:

- Reduced communication overhead
- Lower latency
- Improved scalability
- Reduced power consumption

4.8 LoRa Communication

LoRa communication is used for long-range and low-power wireless transmission between distributed sensor nodes and the monitoring unit.

Advantages of LoRa communication include:

- Long communication range
- Low power consumption
- Internet-independent operation
- Reliable communication in large public spaces

Each node transmits compact data packets containing:

Node ID | CII | Risk Level | Timestamp

4.9 Multi-Node Spatial Correlation

The monitoring unit analyzes risk information received from neighboring sensor nodes to identify localized crowd surges and anomaly propagation patterns.

Sequential anomaly activation across adjacent nodes may indicate directional crowd movement and surge propagation.

This distributed sensing approach improves situational awareness and enables early identification of dangerous crowd behavior across larger monitored regions.

4.10 Alert System

The alert system generates warning notifications based on the computed Crowd Instability Index and adaptive threshold analysis.

Alert conditions include:

- Normal Condition – Green indication
- High-Risk Condition – Warning alert
- Critical Condition – Emergency buzzer and red indication

The alert mechanism enables rapid response during abnormal crowd conditions.

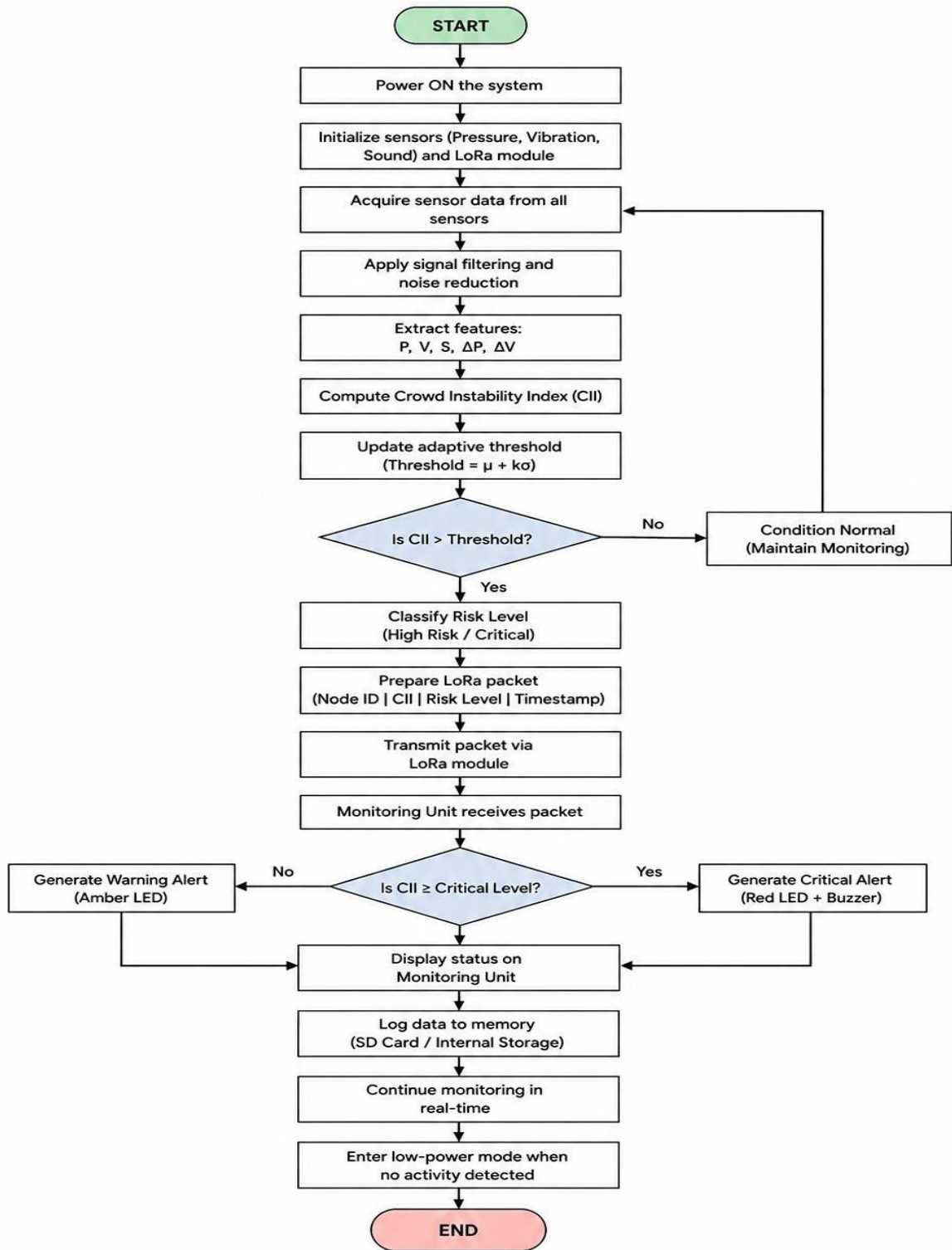


Fig 4.11:Overall Flowchart

4.1.1 Bill of Materials

Table 4.4 :Bill of Materials

S. No.	Component Category	Component / Specification	Quantity	Approx. Unit Cost (INR)	Total Cost (INR)
1	Microcontroller	ESP32 DevKit V1	3	500	1500
2	Communication Module	SX1278 LoRa Module (Ra-02)	1	600	600
3	Pressure Sensing	5kg Load Cell	2	250	500
4	Signal Conditioning	HX711 Load Cell Amplifier Module	2	120	240
5	Motion Sensing	ADXL345 3-Axis Accelerometer	2	350	700
6	Audio Sensing	MAX9814 Microphone Module	2	250	500
7	Gateway Unit	Laptop / PC (Existing System)	1	—	—
8	Connecting Accessories	Breadboard, Jumper Wires, Connectors	1 Set	400	400
9	Mounting Hardware	Screws, Nuts, Fasteners	1 Set	200	200

Total Cost of project: ₹ 4640

Chapter 5:

Results & Discussion

5.1 Controlled Testing Scenarios

Controlled experimental scenarios are created to safely emulate crowd behavior under different conditions. Load variations simulate crowd density buildup, while vibration and sound disturbances emulate panic-induced crowd instability.

The following testing scenarios are considered:

Table 5.1: Controlled testing scenarios

Scenario	Method	Expected Output
Normal Crowd Condition	Light movement and ambient sound	Stable low CII
High Density Condition	Gradual increase in applied load	Increased pressure readings
Panic Surge Condition	Sudden vibration and movement	High vibration energy and rising CII
Noise Burst Condition	Loud sound near microphone	Sound intensity spike
Combined Disturbance	Simultaneous load, vibration, and sound spikes	Critical risk alert

5.2 Mechanical Simulation Setup

A controlled laboratory-scale setup is used to safely emulate crowd instability conditions during testing.

Controlled vibration disturbances, load variations, and sound spikes are introduced individually and simultaneously to validate sensor fusion performance and adaptive threshold response.

This enables repeatable experimentation without requiring actual high-density crowds.

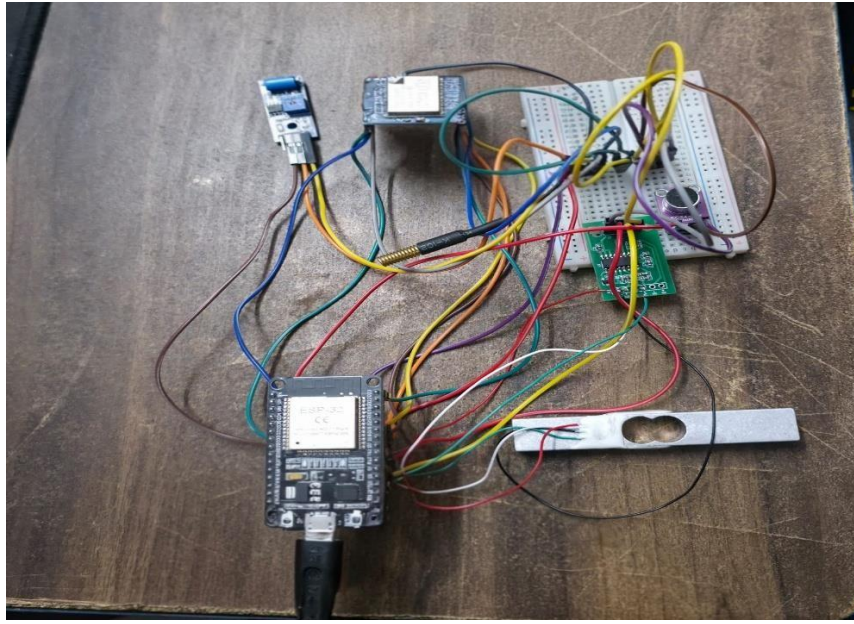


Fig 5.1: Crowd detector node

5.3 Performance Metrics

The proposed system is evaluated using the following performance metrics:

Table 5.2: Performance Metrics

Metric	Purpose
Detection Latency	Measures response speed
False Alarm Rate	Evaluates system reliability
Packet Delivery Ratio	Measures LoRa communication stability
System Stability	Evaluates continuous operation reliability
Response Consistency	Measures repeatability across trials

5.4 System Validation

The system is validated by comparing sensor responses under normal and abnormal simulated crowd conditions.

Sustained increases in pressure, vibration energy, and sound intensity are expected to produce elevated Crowd Instability Index values and trigger adaptive threshold-based alerts.

The distributed sensing architecture successfully identifies sustained abnormal crowd conditions while maintaining low communication overhead and real-time responsiveness.

5.5 Discussion of Results

Experimental analysis indicates that combined pressure buildup, abnormal vibration energy, and panic-induced sound spikes produce significantly higher Crowd Instability Index values compared to normal crowd conditions.

Adaptive threshold analysis effectively reduces false alarms caused by transient disturbances and environmental noise. The use of localized edge processing and LoRa communication enables reliable distributed monitoring with low communication overhead.

The proposed system demonstrates suitability for real-time crowd instability monitoring and early-warning applications in high-density public environments.

```
=====
Weight Level: 0.230
Noise Level: 0.140
Vibration Level: 0.170
Crowd Surge Index: 0.184
d(CSI)/dt: -0.013
LoRa Packet Sent Successfully!
Transmitted Message: CSI=0.184,Noise=0.140,Weight=0.230,Vibration=0.170
=====
```

Figure 5.2: Transmitter node output

```
Packet Received
CSI=0.224,Noise=0.250,Weight=0.310,Vibration=0.050
```

Figure:5.3: Reciever node Output

5.6 Observed Limitations

The proposed system was tested under controlled laboratory-scale conditions. Large-scale real-world crowd validation could not be performed due to safety and infrastructure limitations.

The following limitations were observed during implementation:

- Sensor calibration sensitivity
- Environmental noise affecting microphone readings
- Limited testing area for distributed node deployment
- Dependence on proper threshold tuning
- Limited real-time crowd emulation capability

Despite these limitations, the proposed system successfully demonstrated distributed sensing, adaptive threshold analysis, and real-time anomaly detection functionality.

5.7 Overall System Performance

The implemented prototype successfully demonstrated real-time monitoring of pressure, vibration, and sound-based crowd indicators using distributed sensor nodes.

The proposed architecture achieved:

- Stable sensor-data acquisition
- Reliable localized signal processing
- Successful Crowd Instability Index computation
- Adaptive threshold-based alert generation
- Stable LoRa-based communication between nodes and monitoring unit

The system demonstrated suitability for low-power and privacy-preserving crowd monitoring applications in high-density public environments.

Chapter 6:

Future Trends and Conclusion

6.1 Conclusion

The proposed system presents a distributed LoRa-based sensor network for real-time crowd instability and stampede risk detection using sensor fusion and adaptive threshold analysis. The system utilizes pressure, vibration, and sound sensing to identify abnormal crowd conditions through localized signal processing and Crowd Instability Index computation. Distributed edge processing and LoRa communication enable low-latency monitoring with reduced communication overhead. The proposed architecture operates without camera surveillance or cloud infrastructure, making it suitable for privacy-preserving and low-power public safety applications in high-density environments. The system demonstrates the feasibility of distributed sensing and adaptive threshold-based anomaly detection for early crowd-instability monitoring and warning generation. The developed prototype was successfully validated under controlled testing conditions using simulated crowd disturbances such as pressure buildup, vibration activity, and sound-intensity variation. The implemented system successfully performed localized signal processing, Crowd Instability Index (CII) computation, adaptive threshold analysis, and LoRa-based wireless risk transmission between distributed sensor nodes and the monitoring unit. Experimental observations confirmed stable communication performance and effective abnormal-condition identification with reduced communication overhead.

6.2 Future Trends

The proposed system can be further improved by integrating larger distributed sensor deployments and centralized monitoring dashboards for large-scale public safety applications.

Future enhancements may include:

- GPS-based node synchronization
- Mobile monitoring applications
- Solar-powered sensor nodes
- Cloud-assisted data logging
- Real-time visualization dashboards
- Large-scale smart-city deployment integration

Sustainability Goals Achieved

Table 6.1:Sustainability Goals Achieved

SDG	Description	Contribution
SDG 3	Good Health and Well-Being	Early detection of dangerous crowd conditions and reduction of crowd-related casualties
SDG 9	Industry, Innovation and Infrastructure	Development of distributed embedded sensing and wireless communication systems
SDG 11	Sustainable Cities and Communities	Enhancement of smart public-safety infrastructure for urban environments

Table 6.1:Sustainability Goals Achieved

Technology Readiness Level

The proposed system is currently at: **TRL 4 – Technology Validated in Laboratory Environment**

The developed prototype has been validated under controlled laboratory-scale testing conditions using simulated crowd disturbances such as pressure buildup, vibration activity, and sound spikes.

The implemented system successfully demonstrated:

- Sensor-data acquisition and processing
- Crowd Instability Index computation
- Adaptive threshold-based anomaly detection
- LoRa-based wireless communication
- Real-time alert generation

Further large-scale field deployment and real-world crowd testing are required for higher technology readiness levels and practical public deployment.

TYPE OF PROJECT

The proposed project is classified under the following technical domains:

- Embedded Systems
- Internet of Things (IoT)
- Wireless Sensor Networks
- Signal Processing
- Real-Time Monitoring Systems
- Public Safety and Smart Infrastructure

The project combines distributed sensing, adaptive threshold analysis, and low-power wireless communication for real-time crowd instability and stampede risk detection applications.

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APPENDIX A:

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